

Program 1

Use the **keras** package.

Train a convolutional neural network able to recognise the classes in the CIFAR-10 dataset.

Visualise the architecture of the model using [the available tools](#).

Plot the training and validation losses, note the final error on the testing set. Create a confusion matrix to visualise the missclasifications. Note your thoughts on why the missclasifications are more prominent for the specific classes in your results.

Export the model.

Grading

- 8 points - Network architecture, training, and testing¹
- 2 points - Graphs and analysis

10 points total.

Additional resources

The `report_example.pdf` at the root of the repository shows the plots that you need to generate.

An intuitive explanation of how convolutional neural networks work is available [in this video](#). I recommend skimming the playlist.

More visualisations are available in [the Keras blog](#).

Work to be submitted

All source code files (`.py`, `.ipynb`),

A `.pdf` document with the visualisations and written analysis,

The exported model file.

If the submission does not include all of the above, it will not be checked.

Do not send archived files, IDE configuration, cache, etc.

¹The grade for this rubric is directly proportional to the final accuracy achieved. No points for <60%, full points for >=85%